🛠️ TECHNICAL SPECIFICATIONS: ANALYTIQ PRO (V.1.3)

Expertise: Software Engineering & Business Data Consulting

Major Decision: Pivot to a "Local-First" architecture with SQLite.

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1. ARCHITECTURAL DECISION RECORD (ADR)

Note for Portfolio: This section explains the strategic shift in technology.

Date	Decision	Rationale (Justification)
25/03/26	**Pivot: PostgreSQL ➔ SQLite**	**MVP Simplification:** Eliminates administrative overhead (servers, networking). **Portability:** Database contained in a single .db file. **Agility:** Faster iterations during local testing phase.
25/03/26	**Multilingual Validation**	Native AR/FR/EN support via Gemini to maximize reach (aligned with Morocco 2030 vision).

2. REVISED TECHNICAL STACK

- **Frontend:** React or Vue.js (Modern Web Interface).
- **Backend:** FastAPI (Python) – Lightweight server.
- **Database: SQLite**.
  - *Format:* Local file analytiq_pro.db.
  - *Advantage:* Easy to backup, version on GitHub, and simple to migrate later.
- **Artificial Intelligence:** Gemini 1.5 Pro API (Mapping & Cleaning Engine).
- **Data Processing:** Pandas (Cleaning) & SQLAlchemy (Python ➔ SQLite ORM).

3. INGESTION & MAPPING LOGIC (PRIORITY MODULE)

Development starts with the "Smart Ingest" module.

1. **Agnostic Ingestion:** User uploads any CSV/Excel file.
2. **Semantic Scan (Gemini):** AI identifies columns and maps them to the 6 valuation matrix categories.
3. **SQLite Persistence:** * System dynamically creates a table in analytiq_pro.db.
    - Data is stored in a structured format upon import.
    - *Added Value:* The SQL generated to create these tables is visible to the user.

4. CLEANING & VALIDATION MODULES

- **Chronological Validation:** AI uses SQL to compare date columns and trigger business logic alerts.
- **Contextual Dialogue:** User is prompted regarding anomalies (Refunds, Outliers) before any modification to the SQLite database.
- **Time-Intelligence:** Automatic detection of date ranges (start/end) and suggestion of granularity (Monthly/Quarterly).

5. VALUATION ENGINE (BACKEND)

- **Excel Matrix Integration:** Formulas from your 6 tabs (Financial, Market, Operations, Intangibles, Human Capital, Risk/ESG) are translated into **Standard SQL Queries**.
- **Dashboard:** SQL query results feed the frontend charts.

6. DEVELOPMENT WORKFLOW IMPACT

- **Phase 1:** Local development with SQLite (.db file).
- **Phase 2:** Cloud Deployment. Transitioning to PostgreSQL will be done by simply updating the "connection string" in the code, without rewriting the core SQL logic.